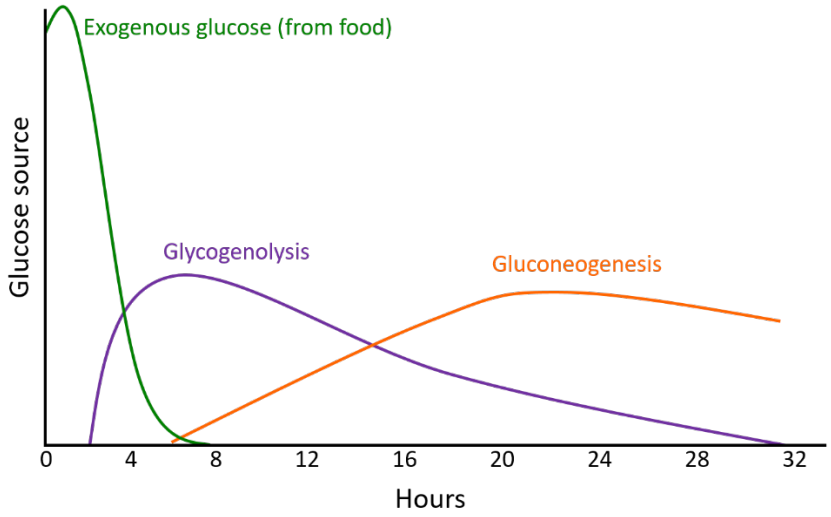
Gluconeogenesis and glycogen metabolism

Gluconeogenesis and **glycogenolysis** (glycogen breakdown) are essential metabolic processes that help **maintain adequate glucose levels in the body**. Some tissues are highly dependent on glucose as a metabolic fuel. Erythrocytes rely exclusively on glucose, whereas the brain uses glucose as its major fuel under normal conditions. In addition, most tissues require glucose for processes such as the synthesis of nucleotides or the synthesis of the carbohydrate portion of glycoproteins and glycolipids. Gluconeogenesis is especially important during **fasting**.



After a meal containing carbohydrates, blood glucose levels rise. Some of the dietary glucose is stored in the liver as glycogen. After 2 or 3 hours of fasting, glycogen begins to be degraded by the process of glycogenolysis, and glucose is released into the blood. As glycogen stores decrease, adipose triacylglycerols start to be degraded, providing fatty acids as an alternative fuel, and glycerol for the synthesis of glucose by gluconeogenesis. Amino acids are also released from the muscle to serve as gluconeogenic precursors. After approximately 30 hours of fasting, liver glycogen stores are mostly depleted and gluconeogenesis becomes the major source of blood glucose.

Gluconeogenesis

Gluconeogenesis is the **synthesis of glucose from noncarbohydrate precursors**. Precursors that can be converted into glucose include lactate, glycerol and most amino acids.

In mammals, **the liver is the major site of gluconeogenesis**. The renal cortex also contributes, particularly during prolonged fasting, and to a lesser extent the intestinal epithelium. In tissues such as muscle and brain, the activity of gluconeogenesis is very low. Hepatic gluconeogenesis contributes to the maintenance of blood glucose levels, ensuring an adequate supply of glucose to glucose-dependent tissues.

Gluconeogenesis and glycolysis share the same endpoints: glucose and pyruvate. However, **gluconeogenesis is not the reverse pathway to glycolysis**. The reason is that the formation of

pyruvate from glucose is a highly exergonic process. Therefore, the simple reversal of glycolysis is not thermodynamically feasible, and a different thermodynamically favorable route is required. Seven of the ten reactions are the same in glycolysis and gluconeogenesis, whereas three steps are different.

Most of the decrease in free energy during glycolysis occurs in three essentially irreversible reactions:

- Glucose + ATP → glucose 6-phosphate + ADP $\Delta G'^{o}$ = -16.7 kJ/mol (catalyzed by *hexokinase*)
- Fructose 6-phosphate + ATP → fructose 1,6-bisphosphate + ADP $\Delta G'^{o}$ = -14.2 kJ/mol (catalyzed by *PFK-1*)
- Phosphoenolpyruvate + ADP → pyruvate + ATP $\Delta G'^{o}$ = -31.4 kJ/mol (catalyzed by *pyruvate kinase*)

Since these three reactions are irreversible, they cannot be used in gluconeogenesis, and these reactions must be bypassed using different enzymes. As a result, both glycolysis and gluconeogenesis proceed as thermodynamically favorable, essentially irreversible pathways.

The three bypass steps of gluconeogenesis are:

- Synthesis of phosphoenolpyruvate from pyruvate
- Formation of fructose 6-phosphate from fructose 1,6-bisphosphate
- Conversion of glucose 6-phosphate into glucose

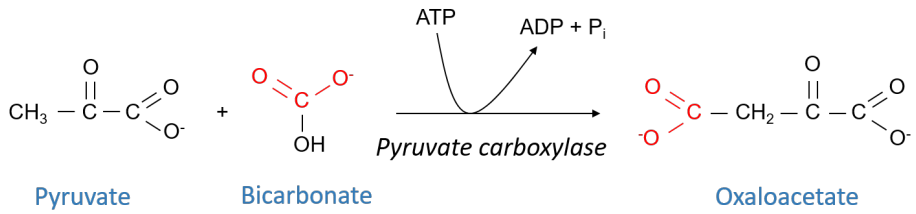
Gluconeogenesis reactions

Synthesis of phosphoenolpyruvate from pyruvate

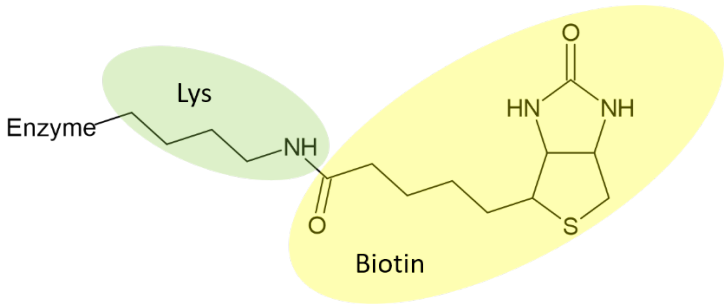
This conversion involves reactions in both the mitochondria and the cytosol and requires two energy-consuming steps.

Pyruvate can be generated by transamination of alanine. It can also be produced from lactate. In this case, the conversion to pyruvate occurs in the cytosol and **pyruvate must be transported into the mitochondria**.

In the mitochondria, **pyruvate is transformed into oxaloacetate**.

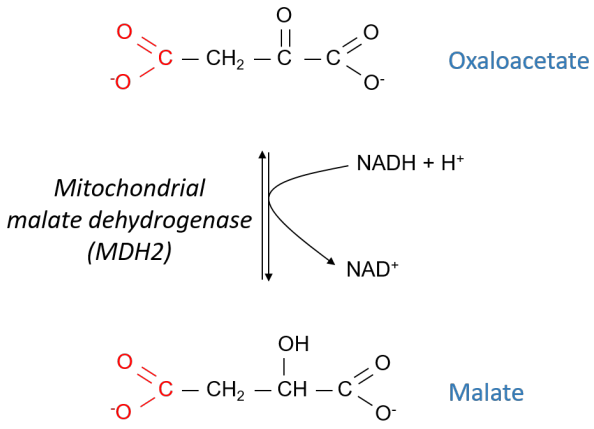


Pyruvate carboxylase, the enzyme that catalyzes the process, contains **biotin** (vitamin H or B_7), a prosthetic group covalently attached to the enzyme through an amide bond with a Lys residue. Biotin plays a key role in many **carboxylation reactions**.



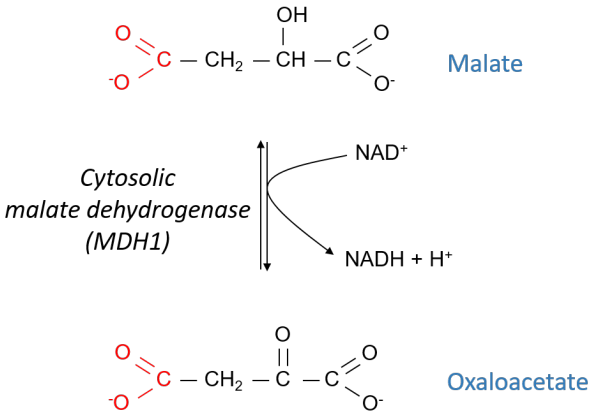
Pyruvate carboxylase is an allosteric enzyme that acts as the first regulator of the gluconeogenic pathway. **Acetyl-CoA is an allosteric activator of pyruvate carboxylase**. High levels of acetyl-CoA, such as those produced during fatty acid oxidation, favor the conversion of pyruvate into oxaloacetate. Oxaloacetate can then be used for gluconeogenesis when cellular energy is abundant or to replenish citric acid cycle intermediates (anaplerotic role of *pyruvate carboxylase*).

If gluconeogenesis continues, oxaloacetate must be transported to the cytosol, where the rest of the reactions will take place. Oxaloacetate cannot cross the inner mitochondrial membrane because there is no specific transporter for it. Therefore, **oxaloacetate is transformed into malate** through the action of ***mitochondrial malate dehydrogenase***.



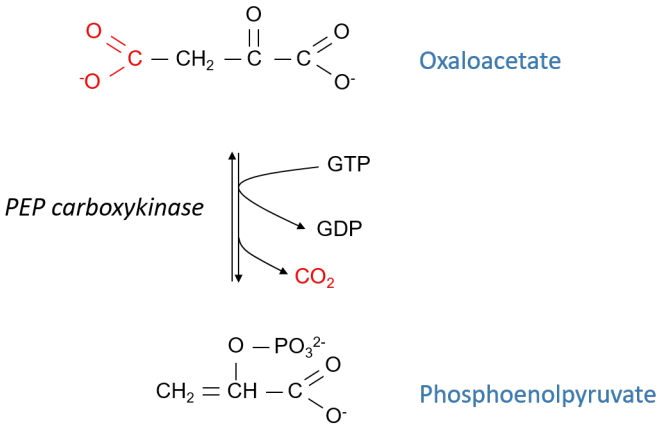
Malate is transported out of the mitochondria through the malate/α-ketoglutarate transporter.

In the cytosol, **malate is reoxidized to oxaloacetate** in a reaction catalyzed by ***cytosolic malate dehydrogenase***.



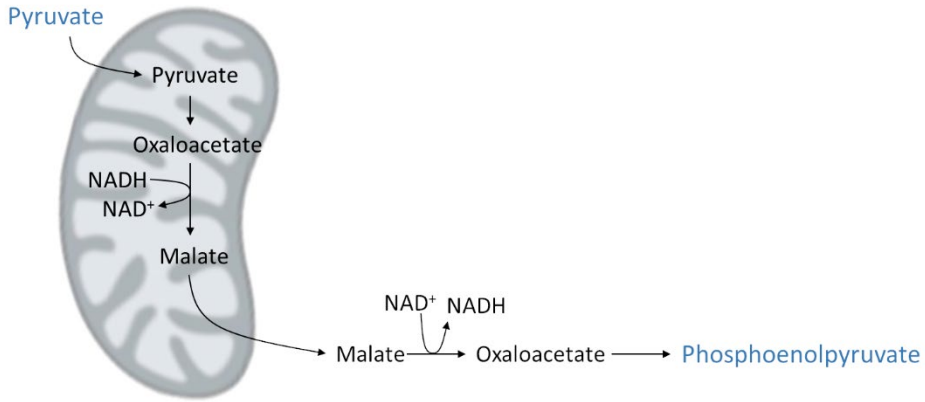
Cytosolic and mitochondrial malate dehydrogenases are examples of **isozymes** (or isoenzymes): different enzymes, encoded by different genes (MDH1 and MDH2, respectively), that catalyze the same chemical reaction.

Finally, **oxaloacetate is decarboxylated and phosphorylated** to produce phosphoenolpyruvate in a reaction catalyzed by ***phosphoenolpyruvate carboxykinase***. The CO_2 incorporated into pyruvate by *pyruvate carboxylase* is subsequently released in this reaction. This carboxylation-decarboxylation sequence represents a way to "activate" pyruvate so that decarboxylation of oxaloacetate allows the synthesis of phosphoenolpyruvate.



In summary, the steps by which pyruvate is converted into phosphoenolpyruvate are:

- Transport of pyruvate from the cytosol to the mitochondria
- Conversion of pyruvate to oxaloacetate
- Reduction of oxaloacetate to malate
- Transport of malate from the mitochondria to the cytosol
- Oxidation of malate to oxaloacetate
- Conversion of oxaloacetate to phosphoenolpyruvate



The **overall balance** of the conversion of pyruvate to phosphoenolpyruvate is:

Pyruvate + ATP + GTP + H_2O → phosphoenolpyruvate + ADP + GDP + P_i + $2H^+$

The consumption of two high-energy phosphate groups allows the process to be thermodynamically favorable. Another important consequence of this process is the transfer of reducing equivalents from the mitochondria to the cytosol, providing the cytosolic NADH required for the reaction catalyzed by *glyceraldehyde 3-phosphate dehydrogenase*.

Synthesis of fructose 1,6-bisphosphate from phosphoenolpyruvate

The next six reactions are the reversible glycolytic reactions operating in the opposite direction:

Phosphoenolpyruvate + H_2O ⇌ 2-phosphoglycerate
(catalyzed by *enolase*)

2-Phosphoglycerate ⇌ 3-phosphoglycerate
(catalyzed by *phosphoglycerate mutase*)

3-Phosphoglycerate + ATP ⇌ 1,3-bisphosphoglycerate + ADP
(catalyzed by *phosphoglycerate kinase*)

1,3-Bisphosphoglycerate + NADH + H^+ ⇌ glyceraldehyde 3-phosphate + NAD^+ + P_i
(catalyzed by *GAPDH*)

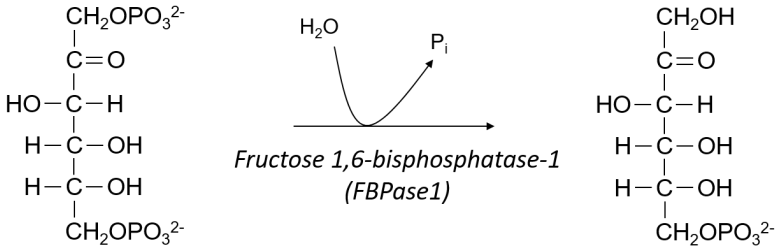
Glyceraldehyde 3-phosphate ⇌ dihydroxyacetone phosphate
(catalyzed by *triose phosphate isomerase*)

Glyceraldehyde 3-phosphate + dihydroxyacetone phosphate ⇌ fructose 1,6-bisphosphate
(catalyzed by *aldolase*)

Note that the reactions from phosphoenolpyruvate to glyceraldehyde 3-phosphate must occur twice, since two three-carbon molecules are required to synthesize one molecule of glucose.

Synthesis of fructose 6-phosphate from fructose 1,6-bisphosphate

This reaction is the second by-pass of the irreversible reaction catalyzed in glycolysis. In gluconeogenesis, **fructose 1,6-bisphosphatase-1** (FBPase1) catalyzes the hydrolysis of fructose 1,6-bisphosphate to fructose 6-phosphate, releasing inorganic phosphate.



Fructose 1,6-bisphosphate — Fructose 6-phosphate

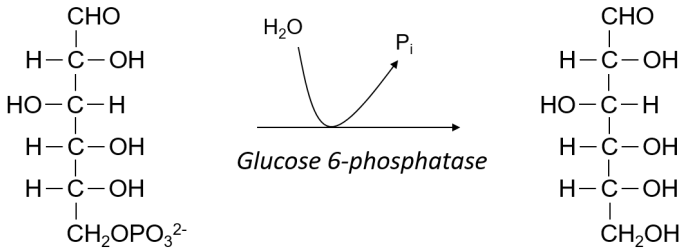
Synthesis of glucose 6-phosphate from fructose 6-phosphate

Fructose 6-phosphate ⇌ glucose 6-phosphate

This isomerization reaction, catalyzed by *phosphohexose isomerase*, is reversible and is shared with glycolysis.

Synthesis of free glucose from glucose 6-phosphate

In the third and final bypass, ***glucose 6-phosphatase*** catalyzes the hydrolysis of glucose 6-phosphate to free glucose and inorganic phosphate. **Glucose 6-phosphatase is expressed mainly in the liver and kidney**, and to a lesser extent in the intestine, but not in skeletal muscle. Therefore, these organs can release free glucose into the bloodstream, whereas muscle glycogen is used locally for energy production.



Glucose 6-phosphate — Glucose

Glucose 6-phosphate enters the endoplasmic reticulum through a specific translocase and its hydrolysis gives rise to free glucose, which is then transported to the cytosol through another specific membrane transporter, and leaves the hepatocyte via the GLUT2 transporter. Free

glucose can then be transported through the bloodstream to other organs, such as the brain and skeletal muscle.

Overall balance of gluconeogenesis

2 Pyruvate + **4 ATP** + **2 GTP** + **2 NADH** + 2 H^+ + 4 H_2O → glucose + 4 ADP + 2 GDP + 6 P_i + 2 NAD^+

Therefore, **gluconeogenesis is a process that requires energy**. This energy expenditure makes gluconeogenesis thermodynamically favorable.

Gluconeogenic precursors

The three major carbon sources for hepatic gluconeogenesis are **lactate**, **glycerol** and **amino acids**. The kidneys use amino acids as major gluconeogenic precursors.

Lactate is produced by erythrocytes and by skeletal muscle during intense exercise. Glycerol is released from adipose stores of triacylglycerols. Glucogenic amino acids come mainly from muscle, where they are obtained by protein degradation. All amino acids except for leucine and lysine are glucogenic because they can be converted to pyruvate or citric acid cycle intermediates. Alanine is the most abundant amino acid delivered to the liver, since it is produced in the muscle from other amino acids.

Important: Humans cannot achieve net glucose synthesis from acetyl-CoA derived from fatty acid degradation.

Regulation of gluconeogenesis

Two steps are the main regulatory points in gluconeogenesis:

1. Pyruvate to phosphoenolpyruvate
2. Fructose 1,6-bisphosphate to fructose 6-phosphate

These steps correspond to reactions in glycolysis that are also catalyzed by regulatory enzymes. The flow of carbon, whether from glucose to pyruvate (glycolysis) or from pyruvate to glucose (gluconeogenesis), depends on the relative activity of the corresponding glycolytic and gluconeogenic enzymes.

Pyruvate to oxaloacetate

Pyruvate is converted to phosphoenolpyruvate when *pyruvate carboxylase* is allosterically activated by high levels of acetyl-CoA (produced, for example, by oxidation of fatty acids). At the same time, ADP, an indicator of low cellular energy status, inhibits *phosphoenolpyruvate carboxykinase*. Thus, when cellular energy is abundant (high acetyl-CoA and low ADP levels), the conversion of pyruvate toward phosphoenolpyruvate is favored.

At the same time, *pyruvate kinase*, the corresponding glycolytic enzyme, is inhibited by PKA-mediated phosphorylation in response to **glucagon**, a hormone released in the pancreas when glucose levels are low. *Pyruvate dehydrogenase*, which catalyzes the conversion of pyruvate into acetyl-CoA, is also strongly inhibited by acetyl-CoA.

In summary, **high levels of acetyl-CoA favor gluconeogenesis and inhibit the oxidation of pyruvate to acetyl-CoA**.

Fructose 1,6-bisphosphate to fructose 6-phosphate

A futile cycle in which both *phosphofructokinase-1* (*PFK1*) and *fructose 1,6-bisphosphatase* (*FBPase1*) act simultaneously is prevented by allosteric regulators with opposite effects on these two enzymes. The key regulator of both processes is **fructose 2,6-bisphosphate**. This regulator strongly activates *PFK-1* and inhibits *FBPase1*. The hormone glucagon induces the degradation of fructose 2,6-bisphosphate. Therefore, **glucagon induces gluconeogenesis and inhibits glycolysis**. Another allosteric regulator at this level is AMP, which activates *PFK-1* and inhibits *FBPase1*, blocking gluconeogenesis when energy levels are low.

Glycogen metabolism

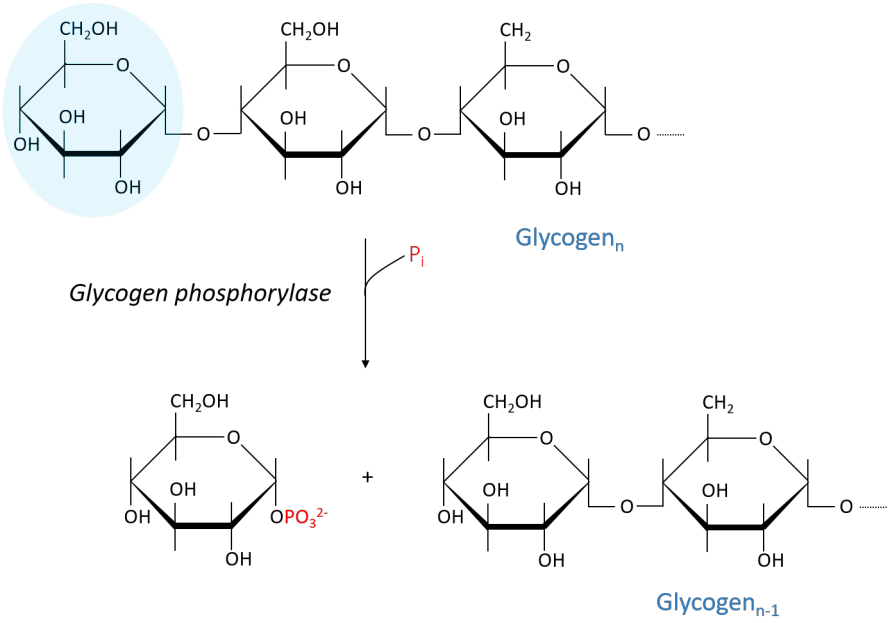
Glycogen is the storage form of glucose in animals. Glycogen is composed of glucose units linked by α1→4 glycosidic bonds, with α1→6 branches occurring with a frequency of approximately one for every ten residues. Glycogen is present in the cytosol in the form of granules. Virtually all cells have the ability to produce glycogen, but the process of glycogen synthesis is especially important in the liver and skeletal muscle. **In the liver, glycogen serves as a glucose reserve that can be mobilized to maintain blood glucose levels, whereas in skeletal muscle it provides glucose 6-phosphate for local energy production**.

As in many metabolic systems, **glycogen synthesis and degradation occur through distinct pathways, allowing them to be independently regulated**.

Glycogen breakdown

Glycogen is degraded to glucose units in a process known as **glycogen breakdown** or **glycogenolysis**.

In the first step, glucose residues are removed at the **non-reducing ends** of the glycogen molecule in a process known as **phosphorolysis**. Glucose residues are released as glucose 1-phosphate. This reaction is catalyzed by ***glycogen phosphorylase***. This enzyme needs **pyridoxal phosphate** (which comes from pyridoxine or vitamin B_6) as a cofactor. *Glycogen debranching enzyme* has two catalytic activities: a transferase activity, which transfers a short segment of glucose residues to another chain, and an α1→6-glucosidase activity, which catalyzes the hydrolysis of the α1→6 bond at the branch point, releasing free glucose. Some degradation of glycogen also occurs within lysosomes, where lysosomal α-glucosidase catalyzes the hydrolysis of glycogen to glucose.



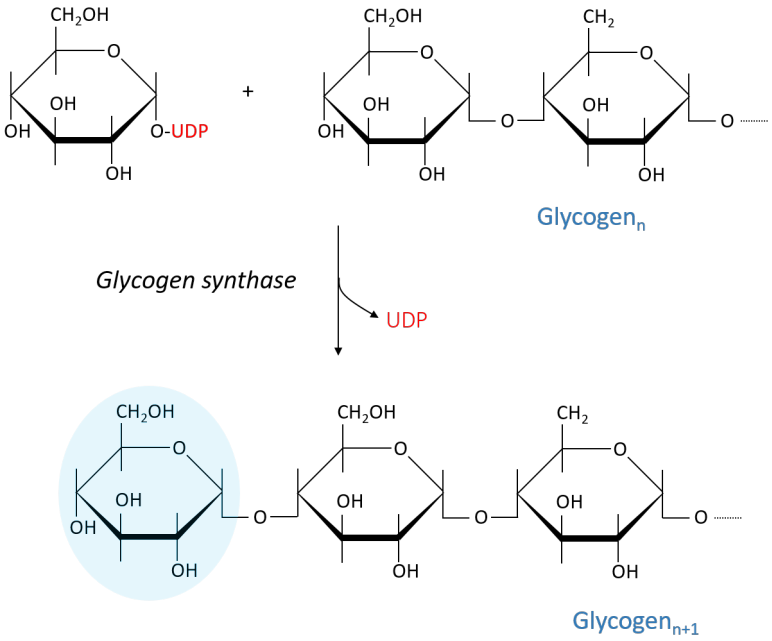
Phosphorolysis is energetically advantageous because the released glucose is already phosphorylated. If free glucose were produced instead, one ATP molecule would be required to convert it into glucose 6-phosphate.

Afterwards, glucose 1-phosphate is isomerized to glucose 6-phosphate in a reaction catalyzed by ***phosphoglucomutase***.

In muscle cells, glucose 6-phosphate is degraded to obtain energy. In liver cells, glucose 6-phosphate enters the endoplasmic reticulum through a specific translocase, where ***glucose 6-phosphatase*** catalyzes its hydrolysis to glucose. Glucose is then transferred to the cytosol through another specific membrane transporter, and leaves the hepatocyte via the GLUT2 transporter. Free glucose can go through the blood to other organs, such as the brain or muscle. Therefore, **the liver is responsible for maintaining a relatively constant level of glucose in the blood**.

Glycogen synthesis

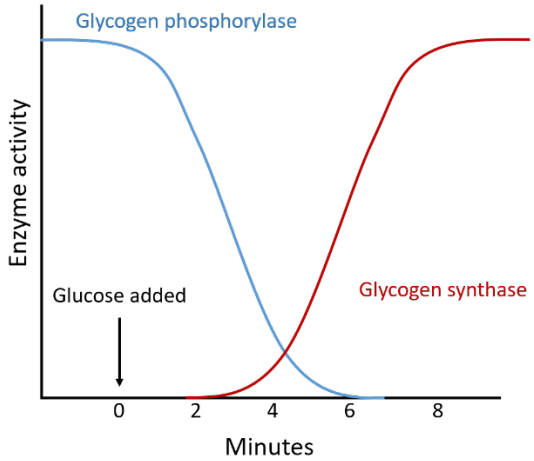
The synthesis of glycogen is catalyzed by ***glycogen synthase***, which requires the use of the nucleotide sugar **UDP-glucose** (uridine diphosphate glucose), an activated form of the monosaccharide. The enzyme catalyzes the transfer of the glucose residue from UDP-glucose to the non-reducing end of the glycogen molecule through the formation of a new α1→4 glycosidic bond. Glycogen synthase requires a primer consisting of a chain of at least eight glucose residues. This primer is generated by ***glycogenin***, which initiates glycogen synthesis. *Glycogen branching enzyme* catalyzes the introduction of branches in glycogen.



The energy required for glycogen synthesis is provided by the activation of glucose as UDP-glucose, which acts as a high-energy glucose donor.

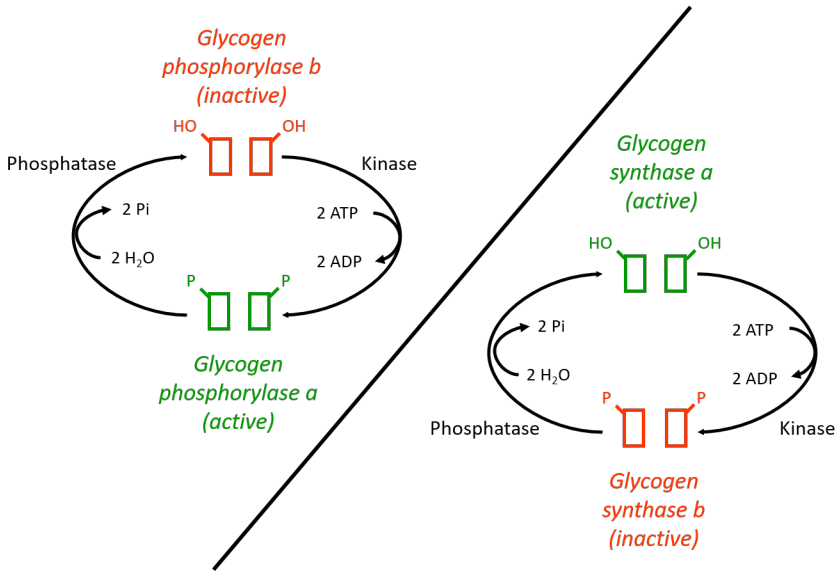
Coordinated regulation of glycogen synthesis and degradation

Glycogen breakdown and synthesis are reciprocally regulated. This prevents glycogen from being synthesized at the same time that it is being broken down.



Adapted from Stalmans W et al. Eur. J. Biochem. (1974)

The main mechanism underlying this coordinated control is the **reversible phosphorylation of *glycogen phosphorylase* and *glycogen synthase***. Phosphorylation activates glycogen phosphorylase but inhibits glycogen synthase. Dephosphorylation has the opposite effects. Thus, mechanisms that promote glycogen breakdown simultaneously inhibit glycogen synthesis, and vice versa.



The active form of glycogen phosphorylase (form a) is phosphorylated. The active form of glycogen synthase (form a) is dephosphorylated.

Hormonal regulation of glycogen metabolism

The main hormones involved in the regulation of glycogen metabolism are insulin, epinephrine and glucagon. These hormones respond to changes in metabolic demand and blood glucose levels and regulate the activity of *glycogen phosphorylase* and *glycogen synthase* in their target tissues. Fasting blood glucose is normally maintained at approximately 80-100 mg/dL.

Insulin is a peptide hormone produced by β cells in the islets of Langerhans of the pancreas. Insulin is released in response to high levels of glucose in the blood. Its overall effect is to lower blood glucose levels. Insulin induces the dephosphorylation of *glycogen phosphorylase* and *glycogen synthase*. Therefore, **insulin inhibits glycogen breakdown and stimulates glycogen synthesis**.

Epinephrine (adrenaline) is a catecholamine released by the medulla of the adrenal glands. Epinephrine is released in response to stress or low levels of glucose in the blood. Epinephrine prepares the body for a fight-or-flight response by rapidly mobilizing energy stores. As part of this response, epinephrine stimulates hepatic glycogenolysis to increase blood glucose and skeletal muscle glycogenolysis to provide glucose 6-phosphate for ATP production. Epinephrine induces the phosphorylation of *glycogen phosphorylase* and *glycogen synthase*. Therefore, **epinephrine stimulates glycogen breakdown and inhibits glycogen synthesis**.

Glucagon is a peptide hormone released by α cells in the islets of Langerhans of the pancreas. Glucagon has the opposite effects of insulin. It is secreted in response to low levels of glucose in the blood. Glucagon induces the phosphorylation of *glycogen phosphorylase* and *glycogen synthase*. Therefore, **glucagon stimulates glycogen breakdown and inhibits glycogen synthesis**.

Glucagon and epinephrine stimulate glycogen breakdown through a phosphorylation cascade. Binding of these hormones to their receptors increases intracellular cAMP levels, which activate PKA. PKA phosphorylates and activates phosphorylase kinase, which in turn phosphorylates

glycogen phosphorylase, converting it into its active form. This cascade amplifies the hormonal signal, allowing a small change in hormone concentration to produce a rapid increase in glycogen breakdown. In skeletal muscle, *phosphorylase kinase* can also be activated by the rise in cytosolic Ca^{2+} that occurs during muscle contraction, directly linking glycogen degradation to the increased demand for ATP.

While both epinephrine and glucagon have similar effects, epinephrine acts on skeletal muscle and the liver, while glucagon acts primarily on the liver, promoting the release of glucose into the bloodstream. Insulin acts on both tissues.

Glycogen storage diseases

A number of genetic diseases are known in which genes encoding enzymes involved in glycogen metabolism are mutated. These diseases are known as **glycogen storage diseases** or **glycogenosis**. Clinical symptoms range from relatively mild to fatal. Most of these diseases affect the storage of glycogen in organs where glycogen accumulates: liver, muscle and, to a lesser extent, kidney. Their frequency is relatively low.

One of the first glycogen storage diseases to be described was **von Gierke disease** (glycogen storage disease type Ia). This disease is associated with mutations in *glucose 6-phosphatase* and it is one of the most serious, since the liver cannot release glucose into the blood. **Glycogen storage disease type 0** is associated with mutations in *glycogen synthase*, which impairs liver glycogen synthesis. In these two diseases, **patients are prone to low blood glucose (hypoglycemia)**. In **Pompe disease** (glycogen storage disease type II) there is an accumulation of glycogen in the lysosomes due to a deficiency of *lysosomal acid α-glucosidase*, affecting multiple tissues, but mainly skeletal and cardiac muscle. In **McArdle disease** (glycogen storage disease type V) there is a deficiency of muscle *phosphorylase*. Patients experience muscle cramps and exercise intolerance, while the liver remains unaffected. Finally, **Lafora disease**, a progressive epilepsy, is caused by mutations in genes that regulate the production of glycogen. The disease is characterized by the formation of Lafora bodies in neurons and other tissues. Lafora bodies contain large amounts of poorly branched glycogen.

Author: Ruben Pio

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